MILITARY SPECIFICATION

METHYL FORMATE
(FOR USE IN AMMUNITION)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade of methyl formate (MF) for use in lithium reserve type batteries. (See 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipping and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC: 6810

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E168 - Infrared Quantitative Analysis, Rec. Practices for
ASTM-E203 - Water Using Karl Fischer Reagent
ASTM-E260 - General Gas Chromatography Procedures, Rec. Practices for
ASTM-E300 - Sampling Industrial Chemicals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Material. The material shall be methyl formate (MF) conforming to the following requirements:

3.1.1 Form. The material shall be a clear colorless liquid at ambient conditions and shall be free of dirt, chips, and other foreign matter when examined as specified in 4.4.3.1.

3.1.2 Infrared Spectroscopy. The material shall conform with the infrared spectrum for methyl formate (see 6.3) when examined as specified in 4.4.3.2.

3.1.3 Purity. The material shall conform to the following when examined as specified in 4.4.3.3.

Impurity	Maximum Limit, ppm
Water	150
Methanol	100
Dimethyl Ether	50
Formic Acid	50
Total Impurities	150

3.2 First article testing. This specification makes provisions for first article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract. (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 2 pounds of MF obtained by sampling as described in 4.4.2.1. The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials, shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.4.3 of this specification.

4.3.3 Rejection. See MIL-A-48078 (PA).

4.4 Quality Conformance Inspection

4.4.1 Lot formation. A lot shall consist of one or more batches of MF produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of MF that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105.

4.4.2 Testing. Sampling and testing shall be performed in a dry atmosphere.

4.4.2.1 Sampling. Approximately 1 liter of MF shall be withdrawn from each batch to be sampled using ASTM Procedure E300-70 for liquids. Samples shall be selected for inspection in accordance with MIL-STD-1235, CSP-1 Plan, Inspection Level II, AQL 6.5%. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed should be tested in accordance with the applicable methods given in paragraph 4.4.3. If any

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of these batches fail to meet any of the test requirements, that batch shall also, be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235. The samples shall be subjected to the tests listed below:

Inspection/Requirement	Defect Classification
Form (See 3.1.1)	Major B
Infrared Spectroscopy (See 3.1.2)	Major B
Purity (See 3.1.3)	Major B

4.4.2.2 Inspection Equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment, shall be employed where applicable for all tests and examinations specified in 4.4.3. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of inspection equipment is required prior to its use for acceptance purpose.

4.4.3 Test methods and procedures. (see 6.2) The tests in 4.4.3.1 thru 4.4.3.3 shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks.

4.4.3.1 Form. Pour sample into a suitable clear glass bottle and visually examine sample for compliance with 3.1.1.

4.4.3.2 Infrared Spectroscopy. Perform an infrared spectrographic analysis of the material in accordance with method given in ASTM-E168.

4.4.3.3 Purity.

4.4.3.3.1 Water. Determine the water content in the sample in accordance with ASTM-E203.

4.4.3.3.2 Other Impurities. Determine the impurities present in the sample using ASTM-E-260 with following additions:

1. Column - Chromosorb 101, 4'x 1/8" diameter
2. Detector - Thermal conductivity
3. Carrier Gas - Helium at 30cc/min

95

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5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A and Level C. Unless otherwise specified, the MF shall be packed in accordance with manufacturer's commercial practice to assure acceptance by commom carrier for safe delivery at first destination for immediate use.

5.2 Marking. Unless otherwise specified, the shipping containers shall be marked in accordance with MIL-STD-129, and shall include the lot number.

6. NOTES

6.1 Ordering data. See MIL-A-48078 (PA).

6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, ATTN: SARPA-QA-A-P, Picatinny Arsenal, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.3 A qualified spectrum of Methyl Formate can be obtained from Sadtler - Research Laboratories - "Sadtler Standard Spectra - Midget Edition", 1956, Infra-Red Spectrogram No. 693.

6.4 Material meeting requirements of this specification were previously obtained from:

Matheson Coleman & Bell
2909 Highland Ave
Norwood, Ohio 45212

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number:6810-AA67

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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